

AMIRUL AZHAJZUL BIN GILOK

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SUMMARY

Systems & Infrastructure Integration Engineer with an MSc background in physical-to-digital systems and registered Graduate Engineer (BEM) credentials. Proven track record in container orchestration (Docker), self-hosted service deployment (TrueNAS SCALE), secure edge ingress (Cloudflare Tunnels), and end-to-end IoT/DAQ telemetry pipelines. Combines Lean Six Sigma root-cause analysis with practical sysadmin experience managing production cloud platforms and hybrid homelab environments. Experienced in scouting, configuring, and maintaining production-grade open-source solutions to deliver high-availability systems without redundant development cycles.

EDUCATION

Master of Science (Mechanical Engineering) Jan 2024 – Sep 2026

Universiti Sains Malaysia - *Viva voce passed (minor corrections)*

- Re-engineered Micro-Perforated Panel (MPP) geometries to eliminate structural redundancies, achieving a 50% spatial footprint reduction while sustaining target performance.
- Validated parametric CAD designs by correlating numerical simulation models against physical testing of 3D-printed prototypes.
- Authored 3 published peer-reviewed research papers and 1 paper under review.

Bachelor of Mechanical Engineering with Honours Sep 2019 – Aug 2023

Universiti Sains Malaysia

- Dean's List Award — 4 semesters.
- Graduate Engineer, registered with the Board of Engineers Malaysia (BEM), Dec 2023.

PROFESSIONAL EXPERIENCE

K-Youth On-the-Job Training (OJT) Trainee Jul 2026 – Present

Khazanah Nasional Berhad [Host: Fortuntech Intelligence Sdn. Bhd, Penang]

- Completed intensive professional readiness training focused on Critical Thinking, Emotional Intelligence, Public Speaking, and Mastering Generative AI Tools for workplace efficiency.
- Transitioning into 3-month OJT at the host company as Field Application Engineer to apply professional upskilling and data-driven methodologies in live operational environments.

Graduate Research Assistant Jan 2024 – Jun 2026

Universiti Sains Malaysia

- Designed and fabricated custom physical test fixtures and prototype structures using 3D printing (additive manufacturing) and SolidWorks, accelerating rapid prototyping and physical validation cycles.
- Led end-to-end experimental validation using specialized Data Acquisition (DAQ) hardware and sensors, correlating empirical measurements with numerical simulation models in MATLAB.
- Formulated standardized testing SOPs and data processing pipelines to maintain measurement accuracy and repeatability across multi-configuration physical trials.

Solar Engineer Trainee Jul 2022 – Oct 2022

Yong Yang Sdn Bhd, Penang

- Supported rooftop PV site surveys and installations, including an institutional project at Universiti Sains Malaysia and several residential sites — verifying mounting/adapter requirements to ensure compliant and safe installations.
- Monitored live system performance via Huawei FusionSolar, comparing generated power output against site weather data to identify discrepancies such as underperformance from soiling, flagging panels that required cleaning schedules.
- Applied foundational electrical checks during design review — verifying systems stayed within inverter capacity limits and assessing how much circuit output would be affected by shading, to reduce avoidable performance loss.

KEY PROJECT

REM-VIAN: IoT-Based Remote Monitoring Vibration Analyzer | Final Year Project

- Designed and built an end-to-end condition-monitoring pipeline — accelerometer sensor > Arduino > Raspberry Pi > cloud dashboard — to detect early machine faults from vibration signatures, enabling remote condition checks from any device instead of relying on physical inspection.

- Validated the system on a test bench using a deliberately unbalanced motor to simulate a real fault condition.
- Awarded Bronze at the International Innovation, Invention & Design Expo (INoDEx 2023), Project ID INO_STE128.

Automated Infrastructure Deployment & Homelab Operations | Personal Project

- Architected and maintained a multi-service hybrid server ecosystem running on TrueNAS SCALE and Docker, hosting decoupled microservices including media platforms (Jellyfin), torrent clients, and automated media ingest pipelines (*arr stack).
- Implemented secure Zero-Trust remote ingress utilizing Cloudflare Tunnels to seamlessly bypass ISP CGNAT restrictions without exposing open inbound firewall ports.
- Managed containerized networking and outbound routing topologies using proxy encapsulation (v2rayA) to maintain uninterrupted data transfer and secure egress paths.
- Configured automated backup schemes, ZFS dataset permissions, and local storage pools for continuous data integrity and multi-client access.

SKILLS

Infrastructure, Containerization & Systems Administration: Docker, Docker Compose, TrueNAS SCALE (ZFS), Linux CLI, container registries (Docker Hub/GitHub), platform deployment and administration (LeConfe), database management (PostgreSQL), and networked storage (Samba/NFS).

Networking, Edge Ingress & Hardware-Software Interfacing: Zero-Trust edge ingress and CGNAT traversal (Cloudflare Tunnels), proxy routing (v2ray), reverse proxies, TCP/IP, and edge DAQ/IoT telemetry pipelines (Arduino, Raspberry Pi, MEMS sensors).

Automation, Diagnostics & Operational Quality: Python, Command Prompt, C/C++ (Arduino IDE), MATLAB, automated data processing, Lean Six Sigma (Yellow Belt), Root Cause Analysis (RCA / 5-Whys), FMEA, and technical SOP documentation.

CERTIFICATIONS

- Lean Six Sigma Yellow Belt — MF Treinamentos (Jan 2026)
- Certified Data Analyst — AntsBees (Jan 2023)
- Graduate Engineer — Board of Engineers Malaysia (Dec 2023)
- Google Project Management Professional Certificate — Coursera (in progress)

LEADERSHIP & EXTRACURRICULAR

System Administrator & Editorial Assistant, ICVSSD 2024 & 2026 **Jun 2024 – Present**

- Deployed and administered the central cloud event management platform (LeConfe), managing secure database privileges, backups, and server stability for hundreds of international authors and reviewers with zero downtime.
- Delivered front-line technical support and rapid incident troubleshooting for authors and reviewers, resolving platform access errors and system queries.
- Streamlined peer-review data pipelines while coordinating on-ground IT infrastructure, network routing, and live technical logistics during on-site conference execution.

Vice President, Postgraduate Student Association (Mechanical Engineering) **Jun 2025 – Jul 2026**

- Directed departmental event pipelines, budgeting, and institutional approval workflows, enforcing project milestone tracking and proactive risk mitigation.
- Demonstrated decisive leadership by identifying low registration early, swiftly restructuring the event into a successful hands-on VR training workshop within one week using existing departmental resources.

Reserve Officer Training Unit (ROTU), Air Force **Sep 2019 – Mar 2024**

- Completed military officer training concurrently with undergraduate engineering studies, subsequently advancing to Training Officer to direct and mentor cadet cohorts. (10 days weekend training each semester, 3-weeks continuous training annually).

LANGUAGES

Bahasa Malaysia (Native), English (Fluent)